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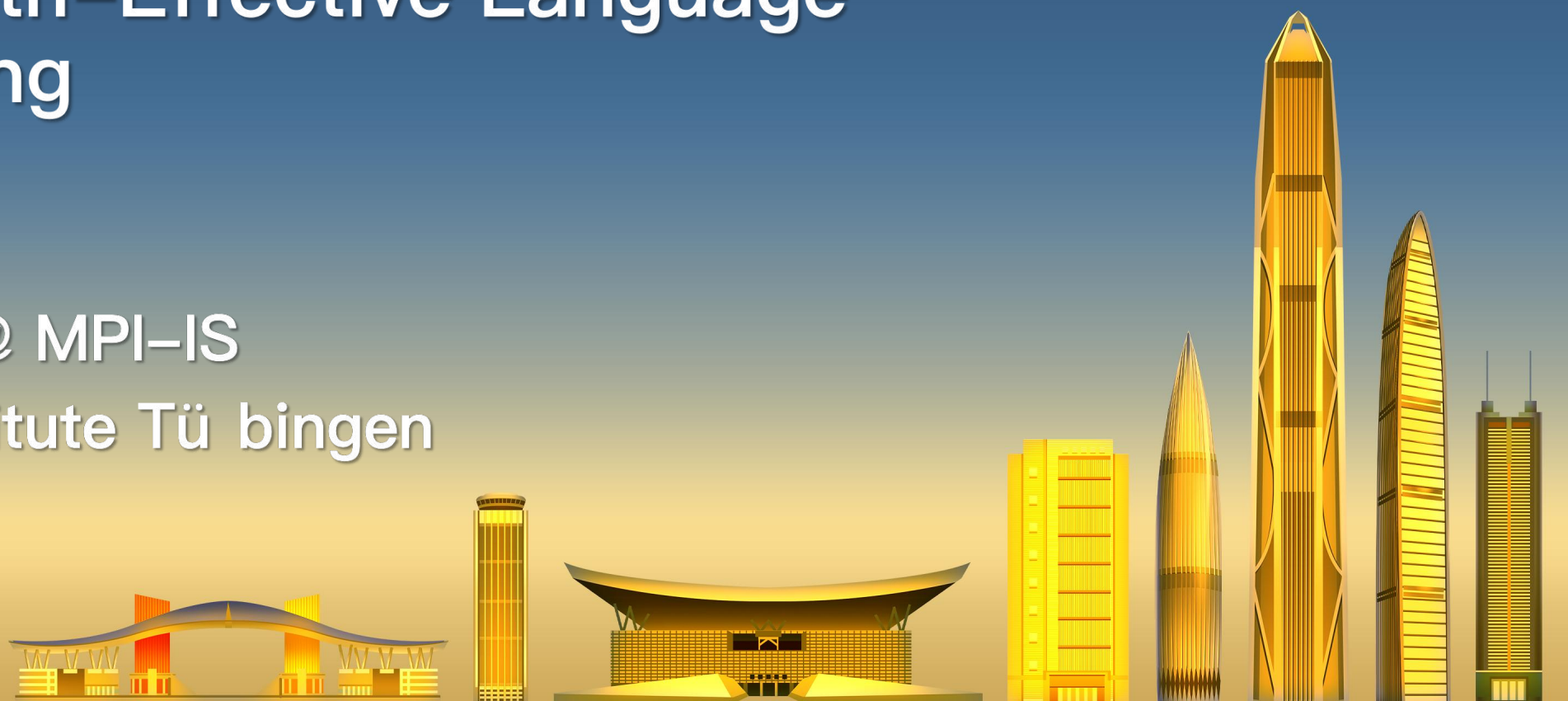
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Rethinking Residual Connections: Toward Depth-Effective Language Model Scaling

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PI @ ELLIS institute Tü bingen



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Part 01 The Curse of Depth in Large Language Models

Part 02 DepthBench: Measuring How Residual Connections Enable More Computational Depth



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01

The Curse of Depth in Large Language Models



Most Deep Layers Can Be Removed in LLMs

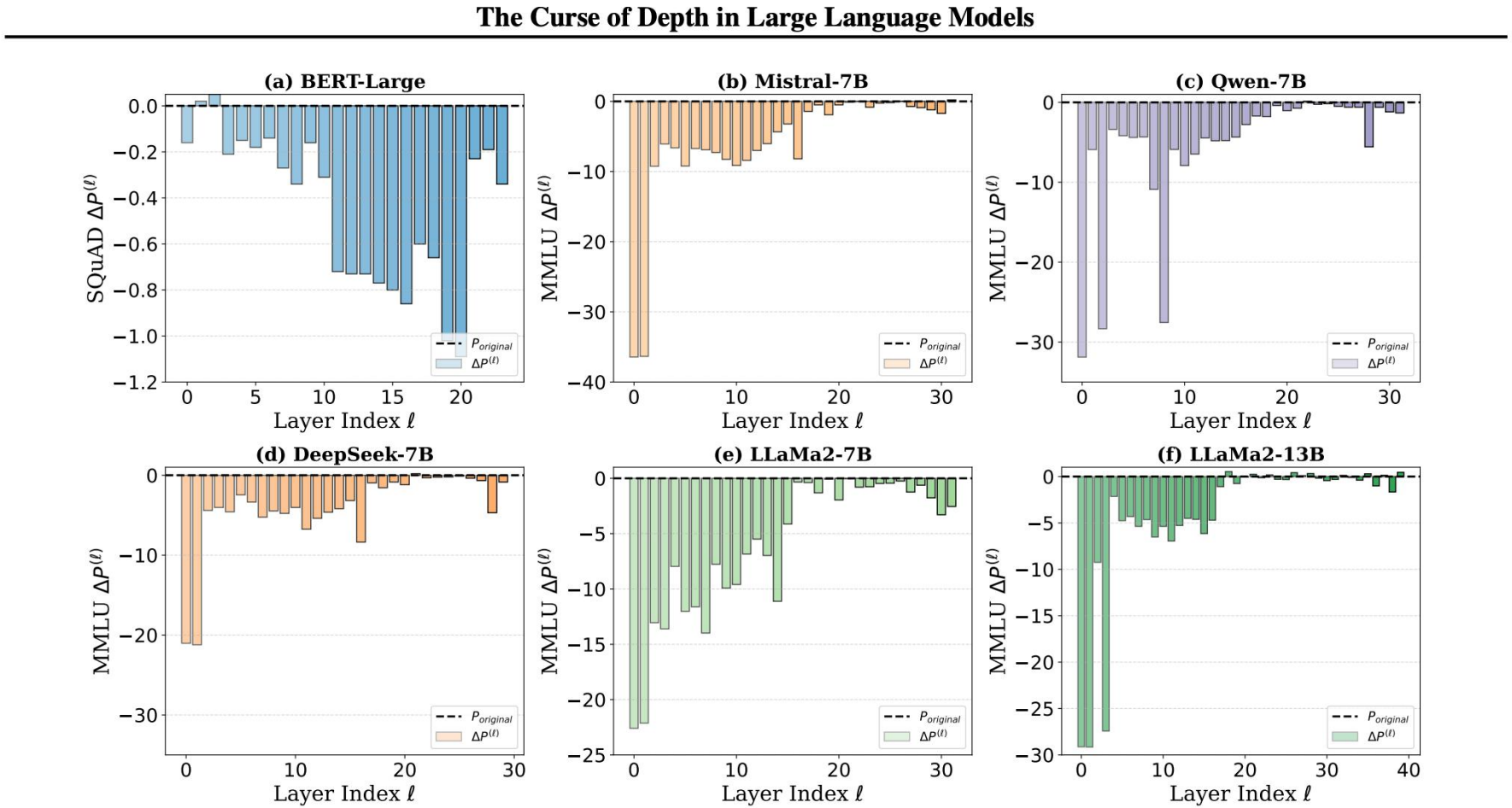
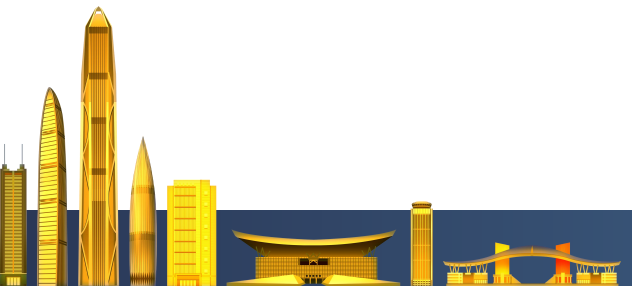


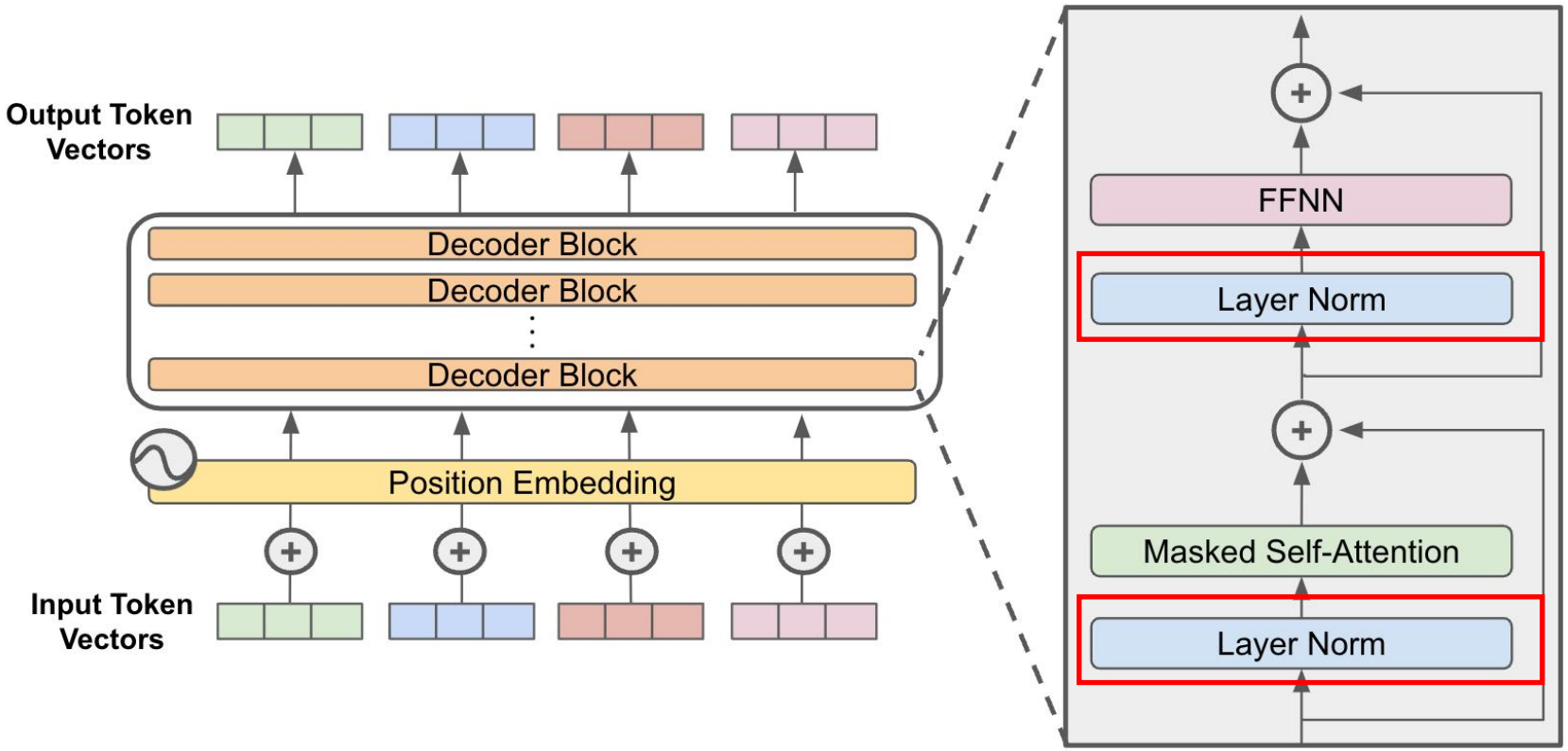
Figure 2. **Performance drop of layer pruning across different LLMs.** (a) BERT-Large (Post-LN), (b) Mistral-7B (Pre-LN), (c) Qwen-7B (Pre-LN), (d) DeepSeek-7B (Pre-LN), (e) LLaMA2-7B (Pre-LN), and (f) LLaMA2-13B (Pre-LN). The results show that Pre-LN models exhibit significant inefficiency in deeper layers, while Post-LN models maintain strong deep-layer contributions.

- **My thoughts:**
  - Ideally, all layers in a model should be well-trained, with sufficient diversity in features from layer to layer, to maximize the utility of the network's parameters



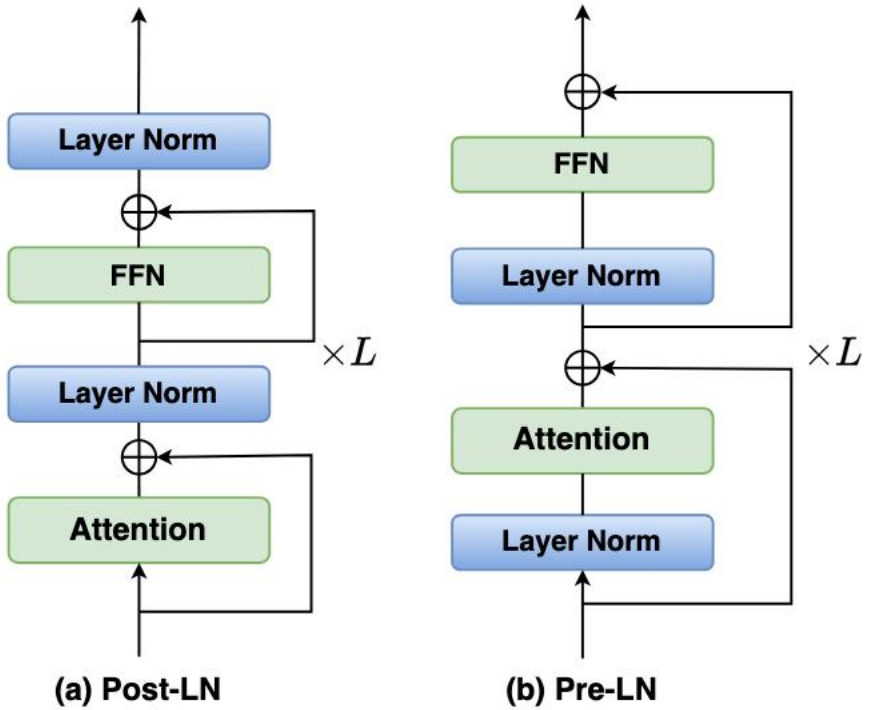
What is the root cause of deep layers' ineffectiveness?

Optimizer?
Architecture?
Attention?
Data?



What is the root cause of deep layers' ineffectiveness?

Transforme
r
BERT
T5



GPT
LLaMa
Mistral

Layer Normalization (LayerNorm) is a key component in transformers, commonly used to stabilize and accelerate training by normalizing the input to each layer of the network. It ensures that the input to a layer maintains a consistent scale and distribution, improving the model's performance and convergence.

Our Hypothesis

The inefficiency of deeper layers in LLMs primarily stems from the choice of Layer Normalization. Specifically,

- **Pre-LN** tends to produce smaller gradients in deeper layers, thereby having **more effective shallower layers**;
- **Post-LN** results in larger gradients in deeper layers but leads to gradient vanishing in earlier ones, thereby having **more effective deeper layers.**



Hypothesis Evaluation (Empirically)

- **Methodology**:
  - **Pre-LN vs Post-LN:** compare layer importance across depths between Pre-LN LLM and Post-LN LLM.
- **Models**:
  - **Open-weight large-scale LLMs**: LLaMa2-7B (Pre-LN) vs. BERT-Large (Post-LN)
  - **In-house small-scale LLMs:** LLaMa-130M (Pre-LN) vs. LLaMa-130M (Post-LN)
- **Layer effectiveness metrics**:
  - **Angular Distance** between adjacent layers
  - **Performance Drop** of layer pruning
  - **Gradient Norm** of each layer

$$d(x^{\ell}, x^{\ell+n}) = \frac{1}{\pi} \arccos\left(\frac{x_T^{\ell} \cdot x_T^{\ell+n}}{\|x_T^{\ell}\| \, \|x_T^{\ell+n}\|}\right) \uparrow$$

$$\Delta P^{(\ell)} = P_{\text{pruned}}^{(\ell)} - P_{\text{original}} \uparrow$$

$$\|\nabla_{\theta}\mathcal{L}(\theta)\|_2 = \sqrt{\sum_i \left(\frac{\partial \mathcal{L}}{\partial \theta_i}\right)^2} \uparrow$$

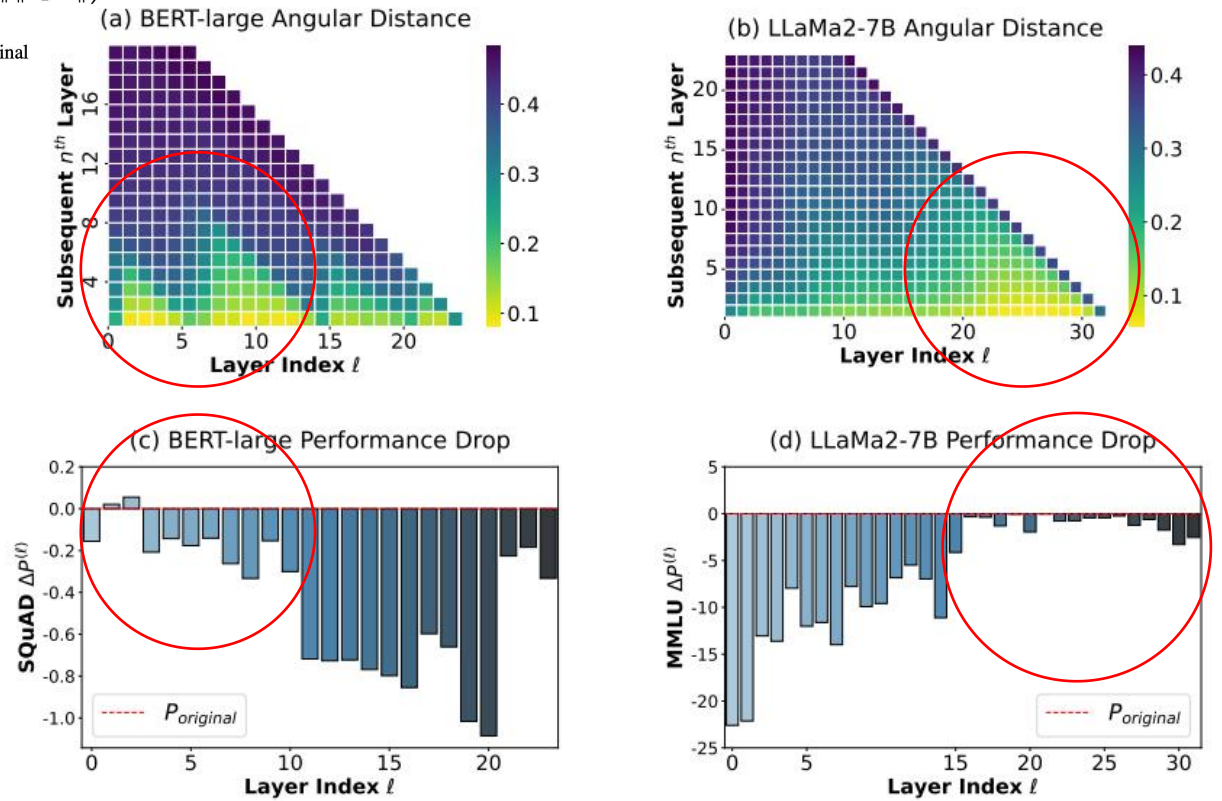
Angular Distance:

$$d(x^{\ell}, x^{\ell+n}) = \frac{1}{\pi} \arccos\left(\frac{x_T^{\ell} \cdot x_T^{\ell+n}}{\|x_T^{\ell}\| \, \|x_T^{\ell+n}\|} \right)$$

Performance Drop:

$$\Delta P^{(\ell)} = P^{(\ell)}_{\text{pruned}} - P_{\text{original}}$$

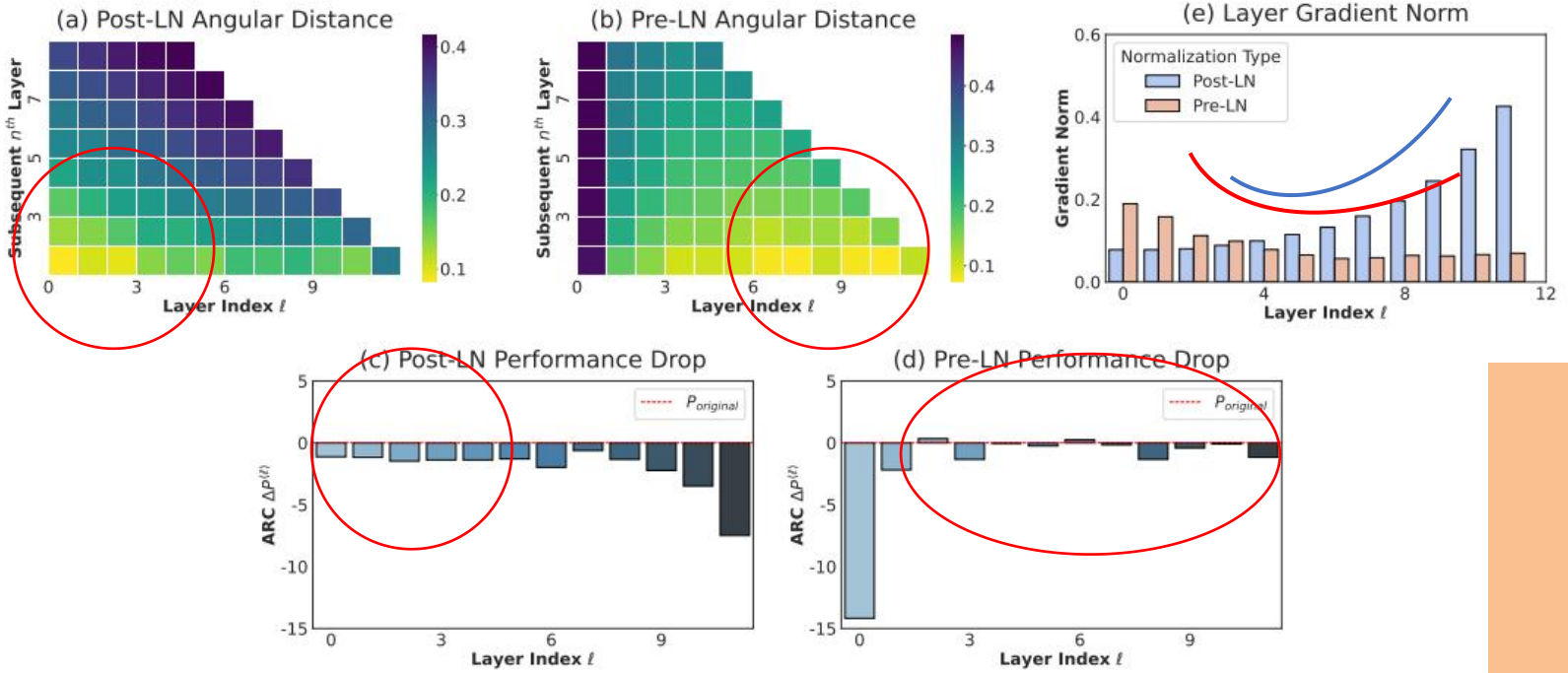
- BERT's early layers are more similar to their adjacent layers (yellow color)
- Pruning entire early layers leads to much smaller performance drop.



- LLaMa2-7B's deeper layers are more similar to their adjacent layers (yellow color)
- Pruning entire deep layers leads to marginal performance drop.

Figure 2: Results of open-weight large-scale LLMs. **Angular Distance (a, b)**: Each column represents the angular distance from the initial layer ℓ (x-axis) and its subsequent n^{th} layer (y-axis). The distance is scaled to the range [0, 1], where yellow indicates smaller distances and purple indicates larger distances. **Performance Drop (c, d)**: (c): SQuAD v1.1 performance drop of removing each single layer from BERT-large; (d): MMLU accuracy drop of removing each single layer from LLaMa2-7B.

- Post-LN leads to larger gradients in deeper layers
- Pre-LN maintains healthy gradient flow in early layers but diminishes in later layers.



- Post-LN's early layers are more similar to their adjacent layers (yellow color)
- Pruning entire early layers leads to much smaller performance drop.

- Pre-LN's deeper layers are more similar to their adjacent layers (yellow color)
- Pruning entire deep layers leads to marginal performance drop.

Figure 3: Results of in-house small-scale LLaMa-130M. **Angular Distance (a, b)**: Each column represents the angular distance from the initial layer ℓ (x-axis) and its subsequent n^{th} layer (y-axis). The distance is scaled to the range [0, 1], where yellow indicates smaller distances and purple indicates larger distances. **Performance Drop (c, d)**: ARC-e performance drop of removing each single layer from LLaMa-130M. **Gradient Norm (e)**: Gradient norm of each layer in LLaMa-130M.

Hypothesis Evaluation (Theoretically)

Post-LN applies $\mathrm{LN}(\cdot)$ after the residual addition:

$$\text{Post-LN}(x) = \mathrm{LN}(x + \mathcal{F}(x)).$$

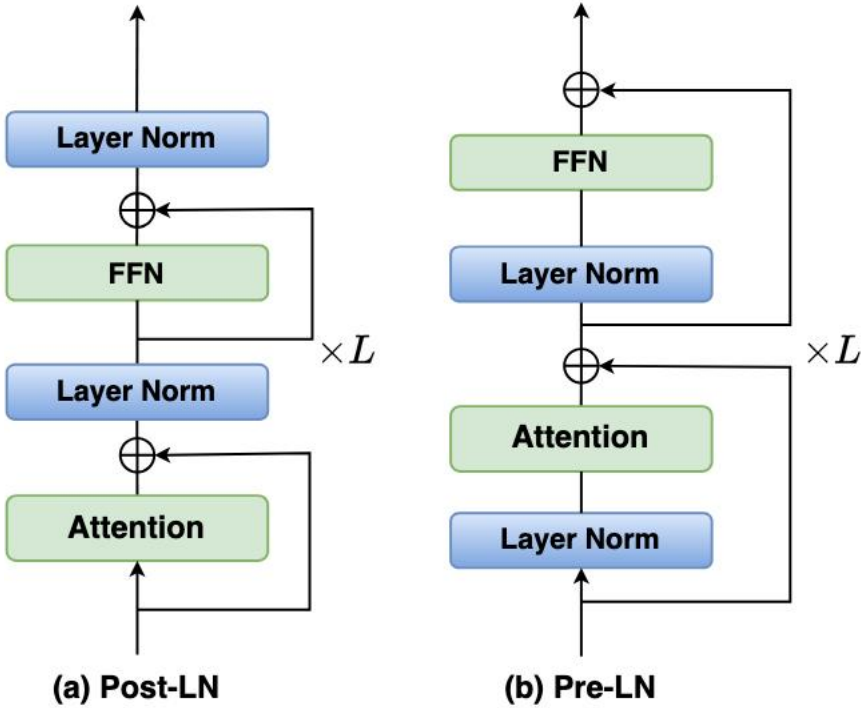
In contrast, Pre-LN applies $\mathrm{LN}(\cdot)$ before the residual addition:

$$\text{Pre-LN}(x) = x + \mathcal{F}(\mathrm{LN}(x)).$$

As shown in our submission, the derivative of Post-LN and Pre-LN are given:

$$\frac{\partial \text{Post-LN}(x)}{\partial x} = \frac{\partial \mathrm{LN}(x + \mathcal{F}(x))}{\partial (x + \mathcal{F}(x))} \left(I + \frac{\partial \mathcal{F}(x)}{\partial x} \right),$$

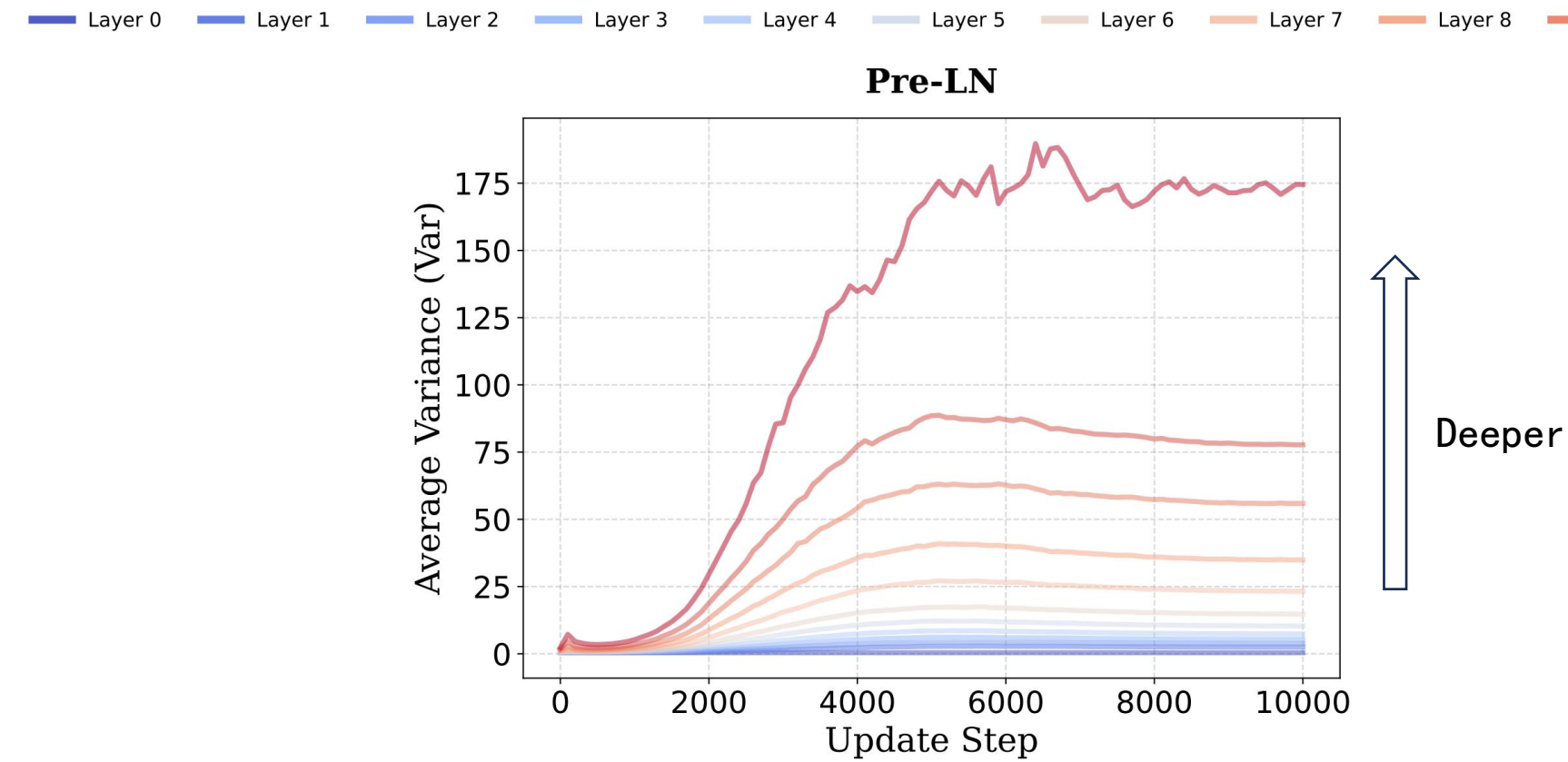
$$\frac{\partial \text{Pre-LN}(x)}{\partial x} = I + \frac{\partial \mathcal{F}(\mathrm{LN}(x))}{\partial \mathrm{LN}(x)} \frac{\partial \mathrm{LN}(x)}{\partial x},$$



Assuming x follows a normal distribution with a mean of 0, σ_x is variance of x, we have

$$\frac{\partial \mathrm{LN}(x)}{\partial x} = \frac{1}{\sigma_x} I \approx 0.$$

Layerwise Output Variance

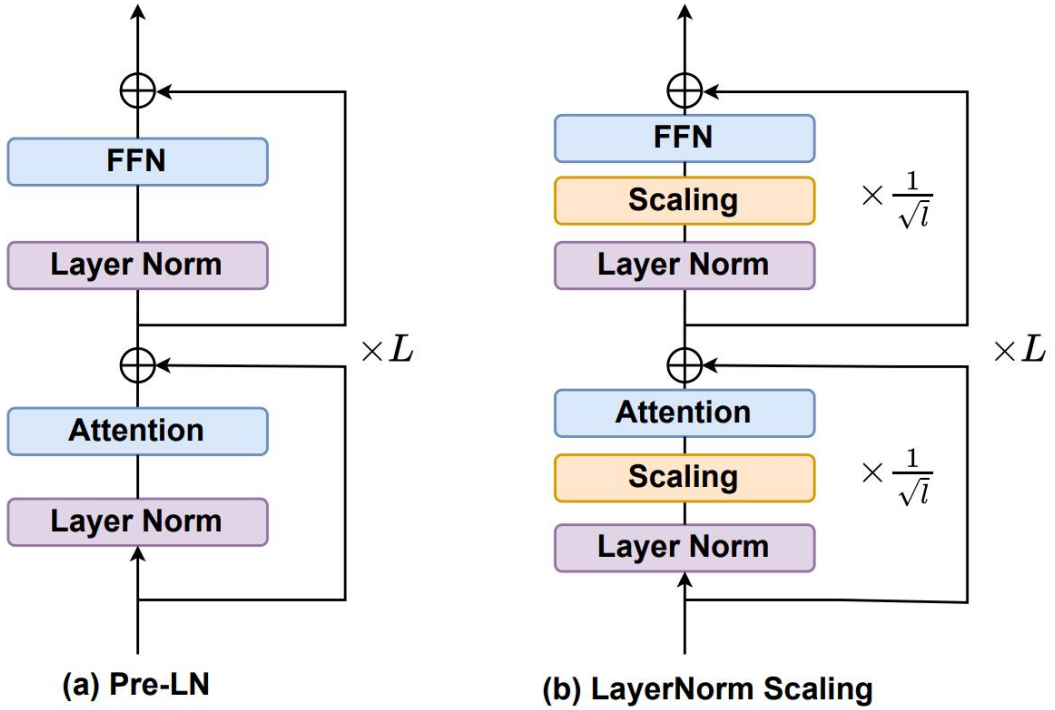


Jacobian of Llama7B (Pre-LN)

Layer 3 Layer 6 Layer 9 Layer 24 Layer 31

The Curse of Depth in Large Language Models

Wenfang Sun [*1] Xinyuan Song [*2] Pengxiang Li [3] Lu Yin [4] Yefeng Zheng [1] Shiwei Liu [5]



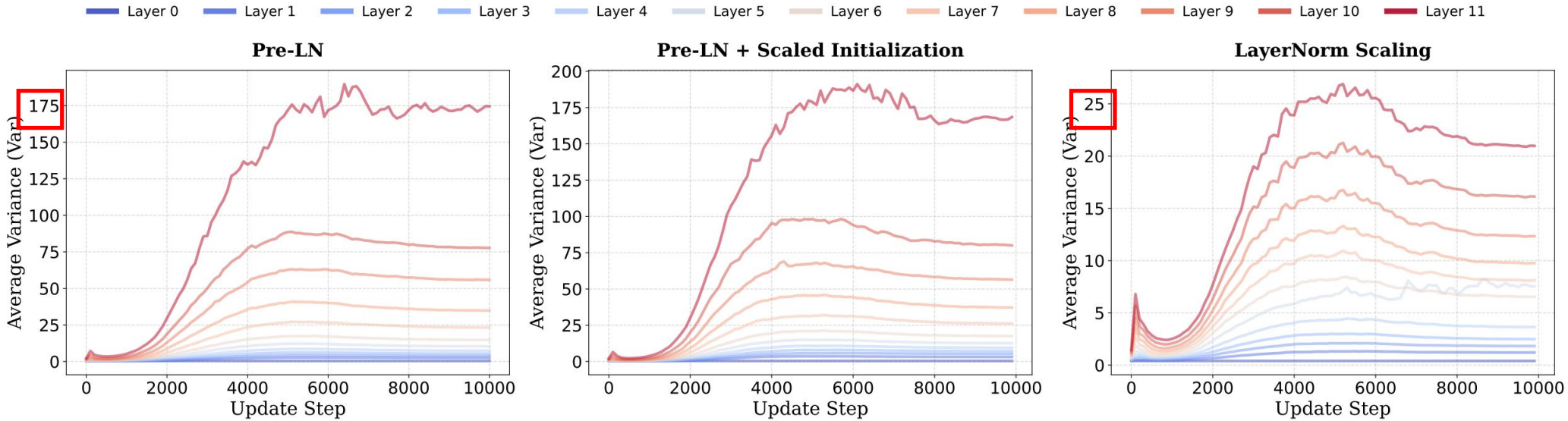


Figure 1. **Layerwise output variance.** This figure compares the output variance across various layers for different setups: (1) Pre-LN; (2) Pre-LN with Scaled Initialization; and (3) LayerNorm Scaling. The experiments are conducted on the LLaM-130M model trained for 10,000 steps. The proposed LayerNorm Scaling effectively controls the variance across layers.

Evaluation Results

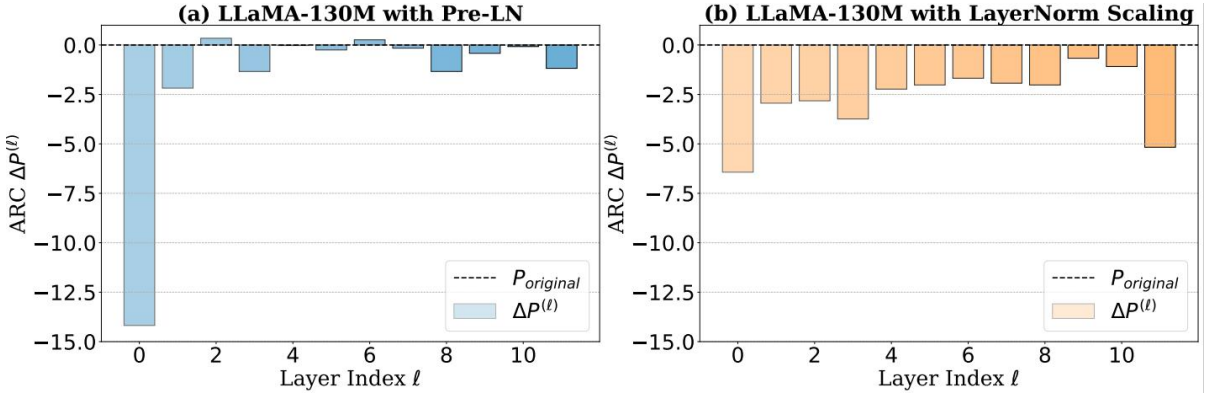
LLM Pre-training Perplexity↓

Table 1. Perplexity (↓) comparison of various layer normalization methods across various LLaMA sizes.

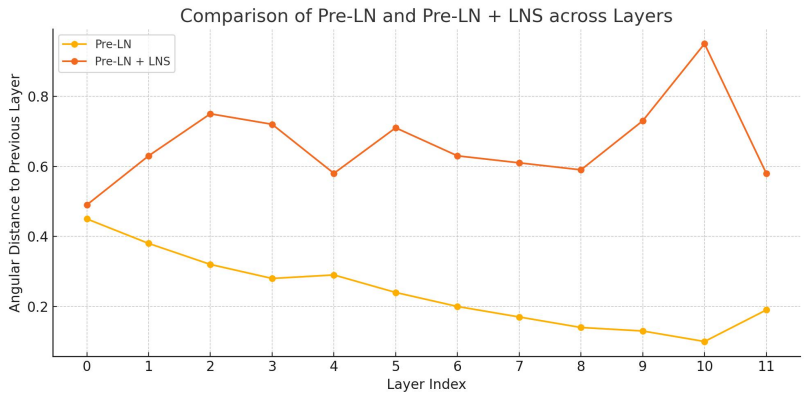
	LLaMA-130M	**LLaMA-250M**	**LLaMA-350M**	**LLaMA-1B**
Training Tokens	2.2B	3.9B	6.0B	8.9B
Post-LN (Ba, 2016)	26.95	1409.79	1368.33	1390.75
DeepNorm (Wang et al., 2024)	27.17	22.77	1362.59	1409.08
Mix-LN (Li et al., 2024b)	26.07	21.39	1363.21	1414.78
Pre-LN (Baevski and Auli, 2019)	26.73	21.92	19.58	17.02
Pre-LN + LayerNorm Scaling	**25.76**	**20.35**	**18.20**	**15.71**

Layerwise Effectiveness Measurement

Performance Drop of Layer Pruning



Layerwise Angular Distance



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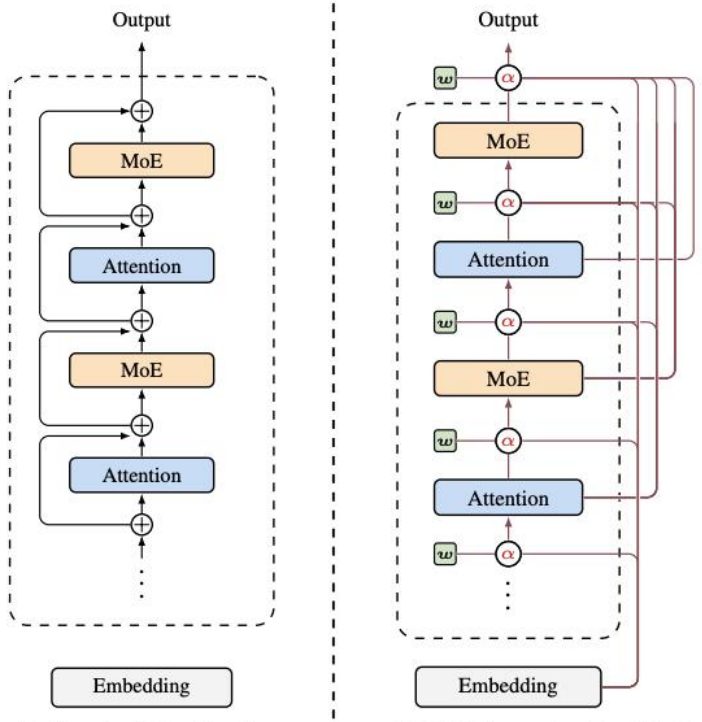
02

DepthBench: Measuring How Residual Connections Enable More Computational Depth



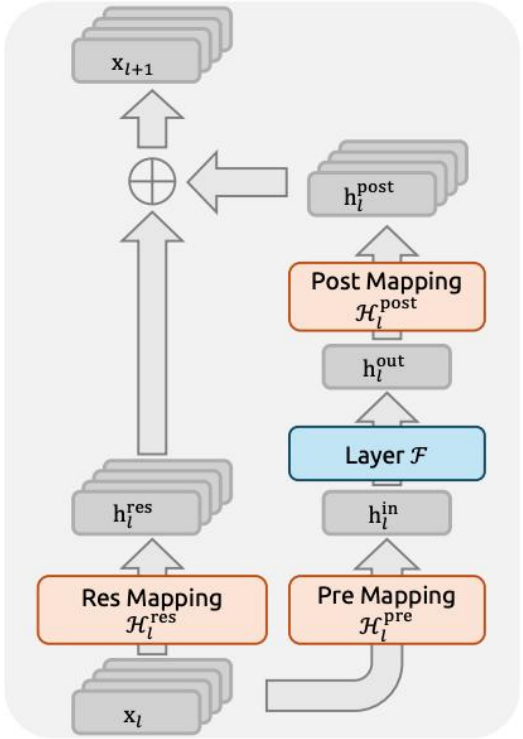
Recent Residual Connections Emerges

KIMI's Attention Residual



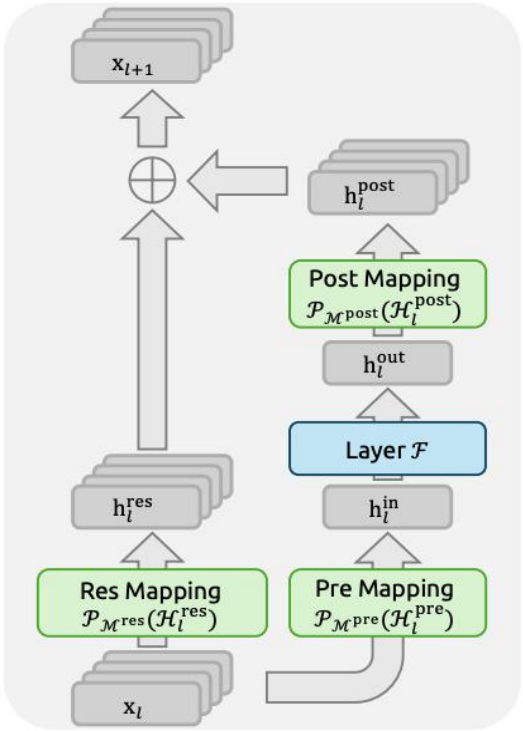
(a) Standard Residuals (b) Full Attention Residuals

ByteDance's HC



(b) Hyper-Connections (HC)

DeepSeek's mHC



(c) Manifold-Constrained HC (*m*HC)

BUT

Which one actually translates increased architectural depth into effective computational depth?
Which one enables greater depth scaling?

DepthBench

Measuring How Residual Connections Enable More *Computational Depth*

Which architectural designs make additional depth computationally effective?
A controlled benchmark of 10 residual designs at a fixed parameter budget.

Keyu Wang [1,2,3*] Yangyi Huang [4*] Jiale Kang [4] David González-Martínez [1,2,3]

Weiyang Liu [4] Shiwei Liu [1,2,3]



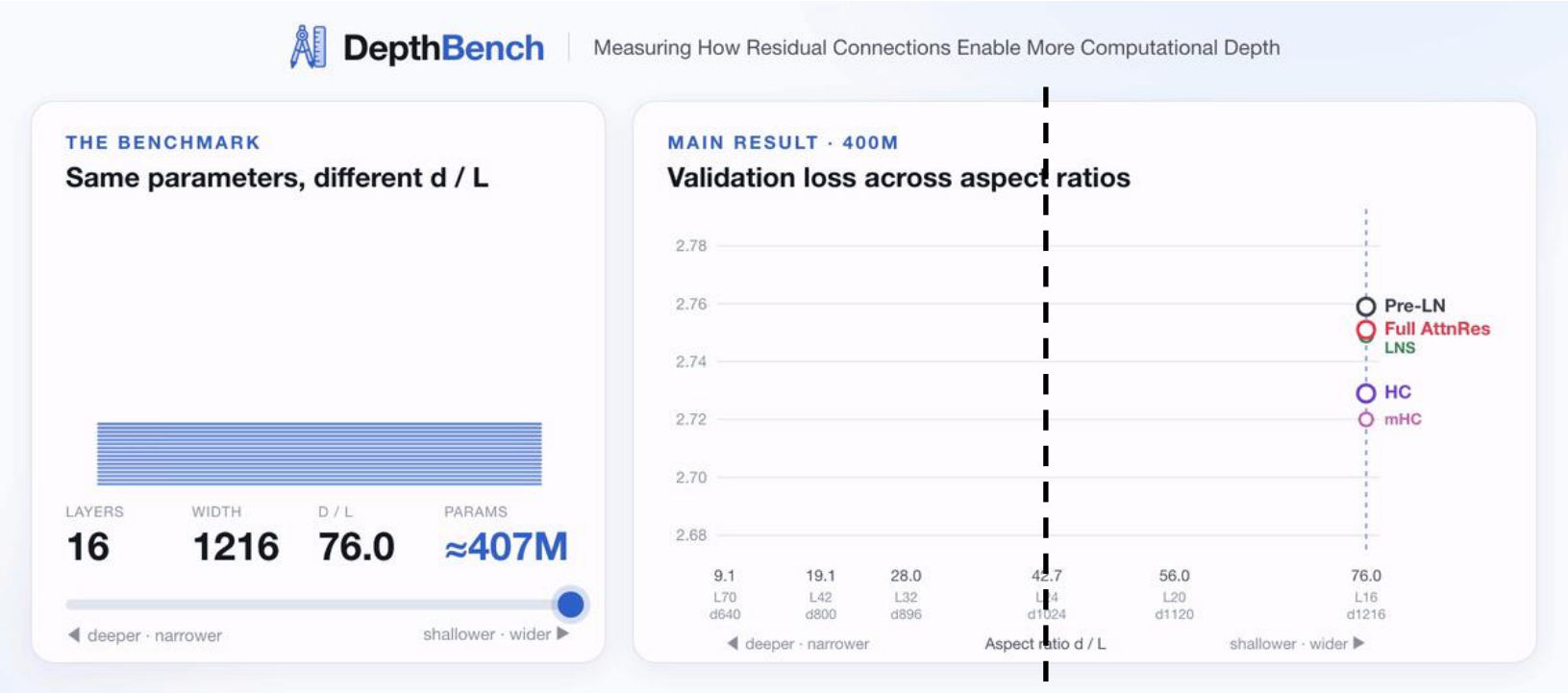
What do we evaluate? 10 residual connections.

Method	Update rule	Weight	Source
Baseline			
Pre-LN (Xiong et al., 2020)	$h_\ell = h_{\ell-1} + F_\ell(\mathrm{LN}(h_{\ell-1}))$	Fixed	$h_{\ell-1}$
LayerNorm variants			
Sandwich-LN (Ding et al., 2021)	$h_\ell = h_{\ell-1} + \mathrm{LN}_{\mathrm{out}}(F_\ell(\mathrm{LN}_{\mathrm{in}}(h_{\ell-1})))$	Fixed	$h_{\ell-1}$
LNS (Sun et al., 2026)	$h_\ell = h_{\ell-1} + F_\ell\left(\frac{1}{\sqrt{\ell}}\mathrm{LN}(h_{\ell-1})\right)$	Fixed	$h_{\ell-1}$
DeepNorm (Wang et al., 2024)	$h_\ell = \mathrm{LN}(\alpha h_{\ell-1} + F_\ell(h_{\ell-1}))$	Fixed	$h_{\ell-1}$
KEEL (Chen & Wei, 2026)	$h_\ell = \mathrm{LN}(\alpha h_{\ell-1} + F_\ell(\mathrm{LN}(h_{\ell-1})))$	Fixed	$h_{\ell-1}$
Multi-stream residuals			
HC (Zhu et al., 2025)	$H_\ell = H_{\ell-1}A_\ell + F_\ell(\mathrm{LN}(H_{\ell-1}\alpha_\ell))\,\beta_\ell^\top$	Dynamic	m residual streams
mHC (Xie et al., 2025)	$H_\ell = H_{\ell-1}\mathrm{Sinkhorn}(\widetilde{A}_\ell) + F_\ell(\mathrm{LN}(H_{\ell-1}\sigma(\widetilde{\alpha}_\ell)))\,2\sigma(\widetilde{\beta}_\ell^\top)$	Dynamic	m residual streams
Cross-layer access			
AttnRes (Team et al., 2026b)	*Full:* $h_\ell \propto \sum_{i=0}^{\ell-1}\phi(w_\ell, v_i)v_i, \quad v_0 = h_1,\ v_{i\geq 1} = f_i(h_i)$	Dynamic	$[h_1, \ldots, h_{\ell-1}]$
	Block: $h_\ell \propto \sum_{i=0}^{n-1}\phi(w_\ell, v_i)v_i + \phi(w_\ell, v_n^j)v_n^j$	Dynamic	$[v_0, \ldots, v_{n-1}, v_n^j]$
MoDA (Pre-LN) (Zhu et al., 2026)	$h_\ell = h_{\ell-1} + \mathrm{MoDA}_\ell(\mathrm{LN}(h_{\ell-1}); \mathcal{C}_{<\ell})$	Dynamic	$\left[h_{\ell-1,\leq t}; h_{0,t}, \ldots, h_{\ell-2,t}\right]$

How do we evaluate?

- Fixed parameter budgets: 400M - 1.6B
- Varied aspect ratio (d_model / n_layer): 9.1 (deep & narrow) to 76.0 (shallow & wide)
- Tuned learning rate separately for each aspect ratio and each method
- Question: Do these methods turn extra depth into lower loss?

Overall Results



KEY FINDINGS

Depth pays off — only with the right residual connection.

01 / BASELINE

Pre-LN does not benefit from depth.

02 / NEW RESIDUALS

HC & Full AttnRes keep improving.

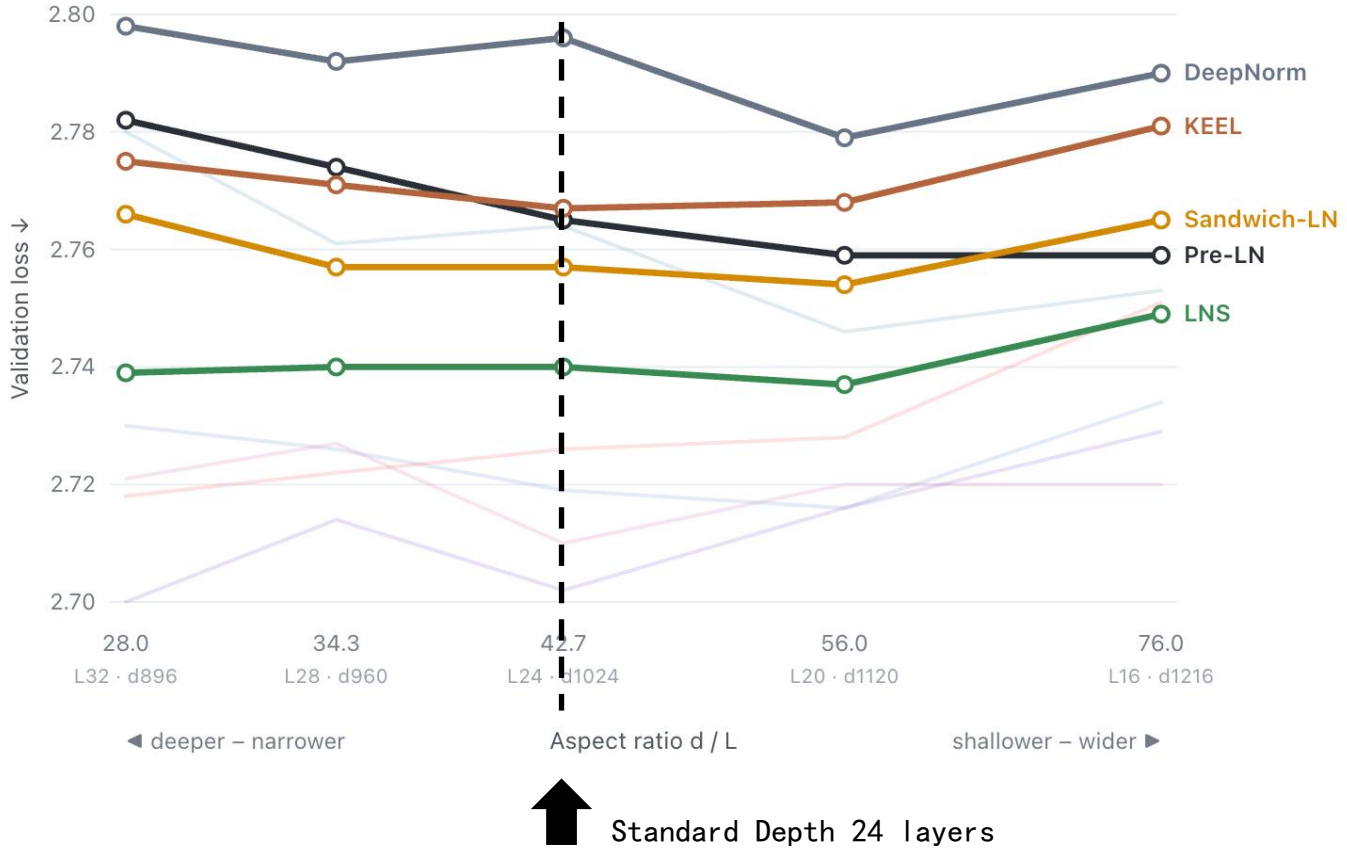
03 / MECHANISM

Residual design changes how depth computes.

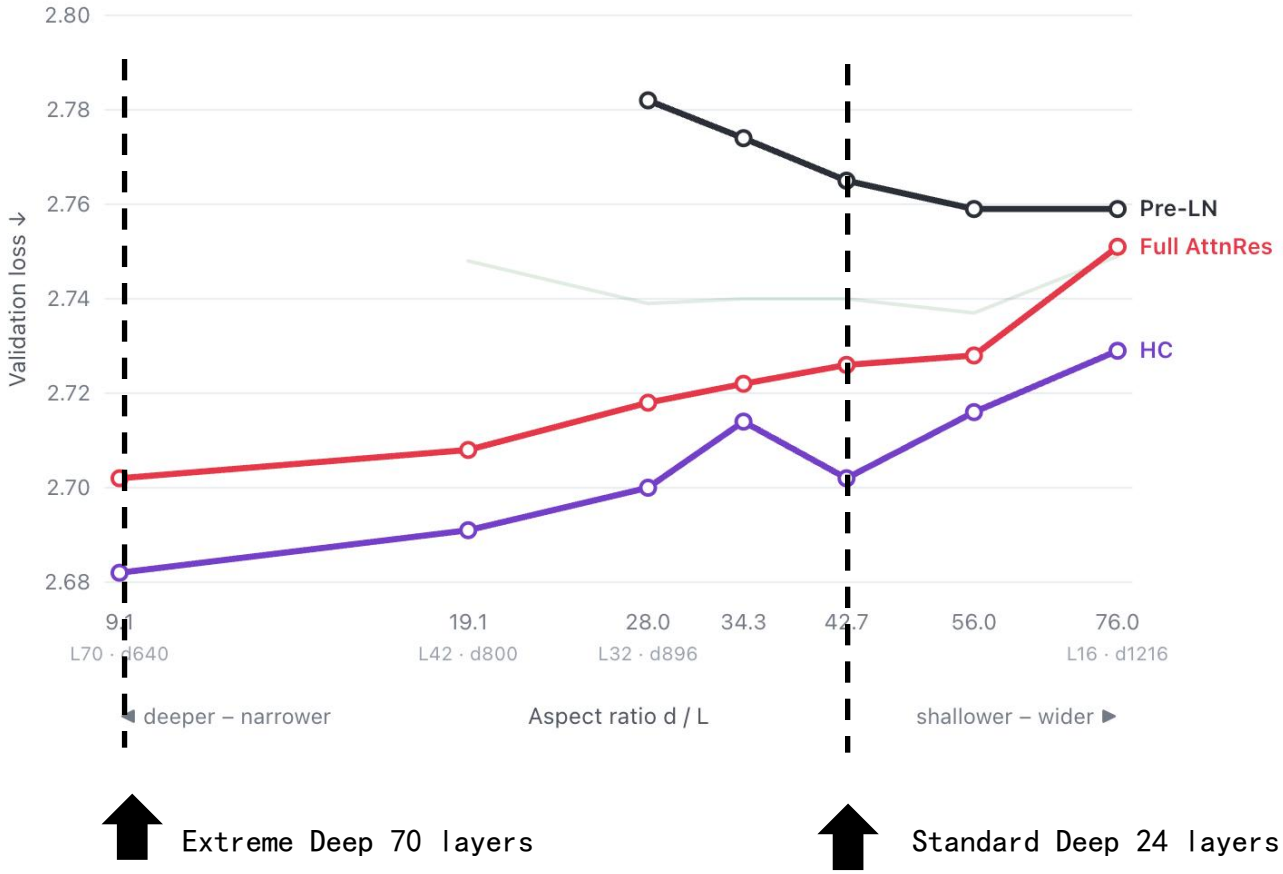
04 / COST

No free lunch.

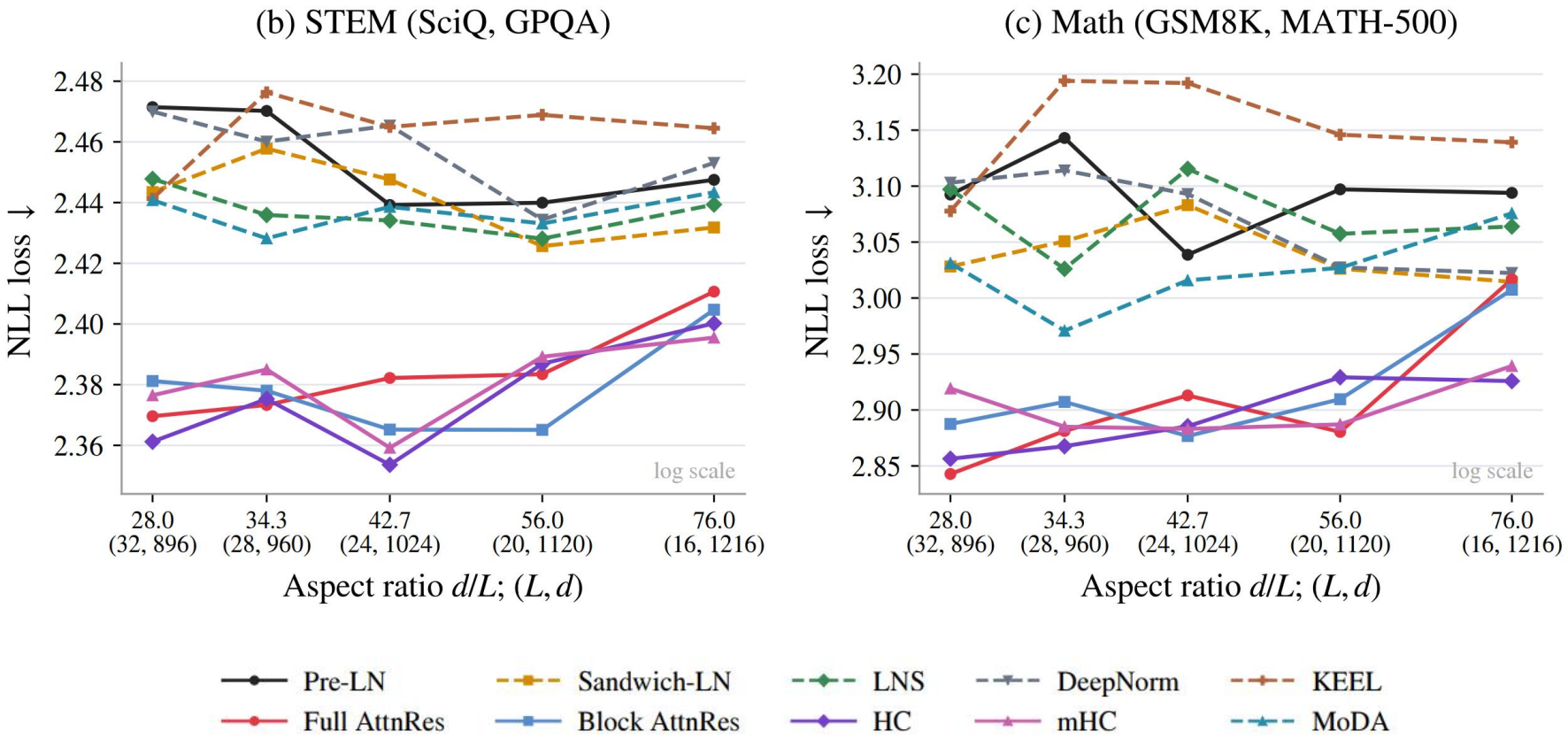
Even at fixed parameters, deeper



1. Conventional residual connections do not benefit consistently from deeper architectures at iso-parameter.
2. Pre-LN prefers wider network. The standard ratio is also too narrow.
3. Sandwich-LN and LNS outperform Pre-LN with more depths, but don’t deliver better loss.



1. New residual designs unleash the power of deeper models even with extremely small aspect ratios.
2. HC and Full AttnRes continue to reduce loss as models become deeper, even at an extreme aspect ratio of 9.



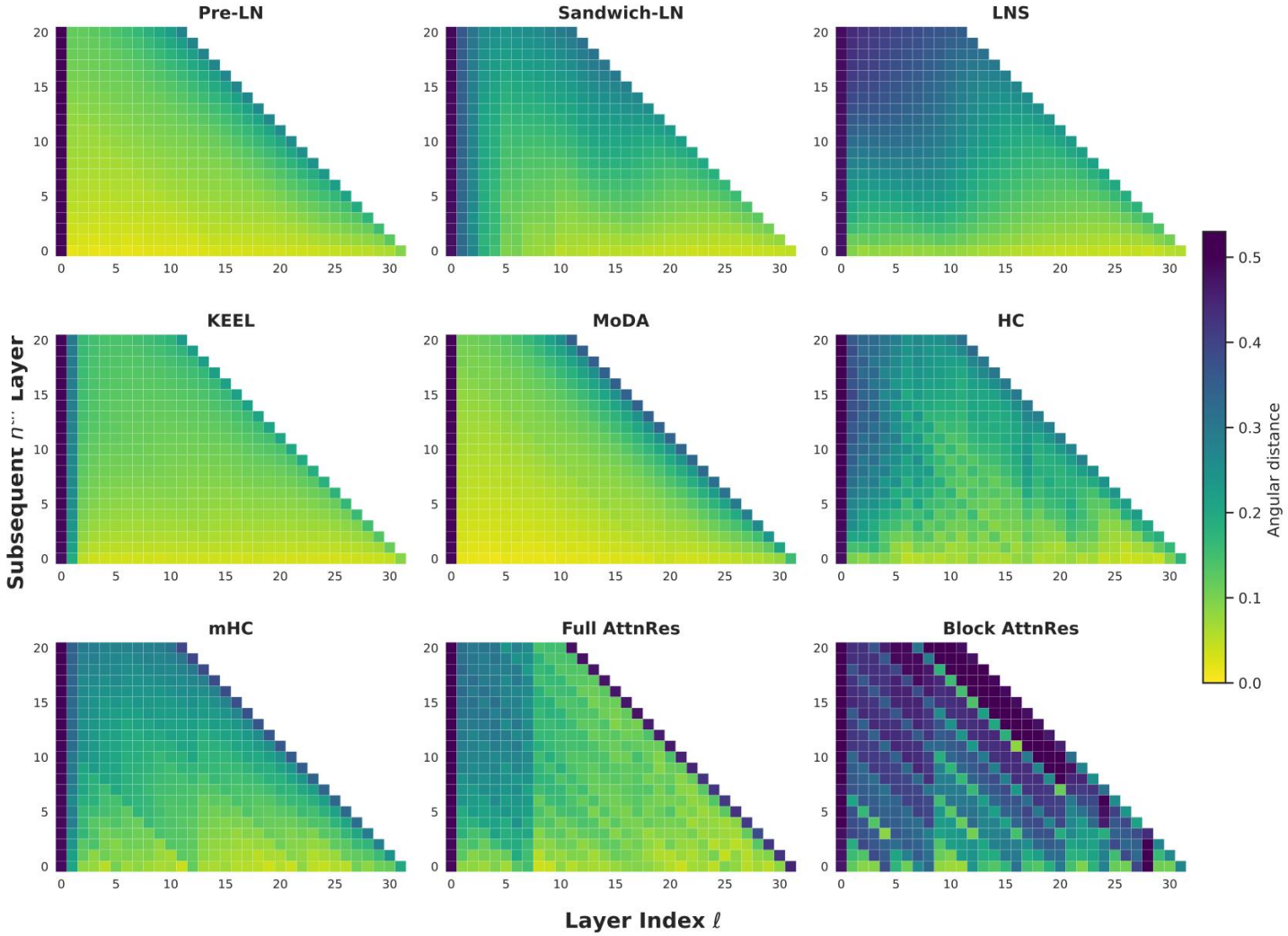
1. The benefits of depth scaling extend beyond pre-training loss.
2. The gains of HC and Full AttnRes transfer to STEM and Math tasks.

Do late layers keep doing distinct work?

Layer Similarity

ANGULAR DISTANCE

$$d(h_\ell, h_{\ell+n}) = \frac{1}{\pi} \arccos\left(\frac{h_\ell \cdot h_{\ell+n}}{\|h_\ell\|_2 \, \|h_{\ell+n}\|_2} \right)$$



Do late layers keep doing distinct work?

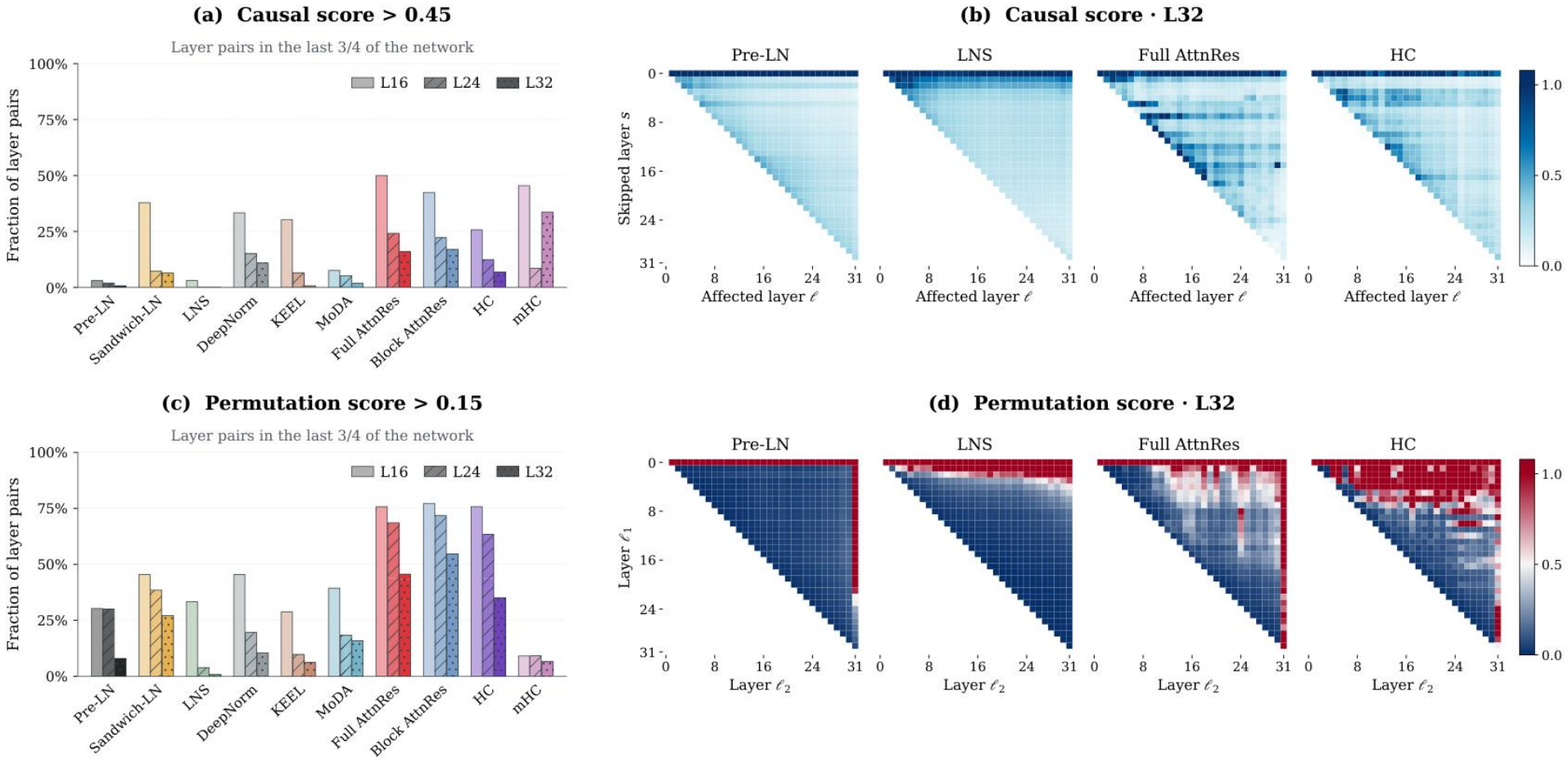
Layer Perturbation

CAUSAL SCORE

$$C(s,\ell) = \frac{\|(h_{\ell+1} - h_\ell) - (\bar{h}_{\ell+1} - \bar{h}_\ell)\|_2}{\|h_{\ell+1} - h_\ell\|_2}$$

PERMUTATION SCORE

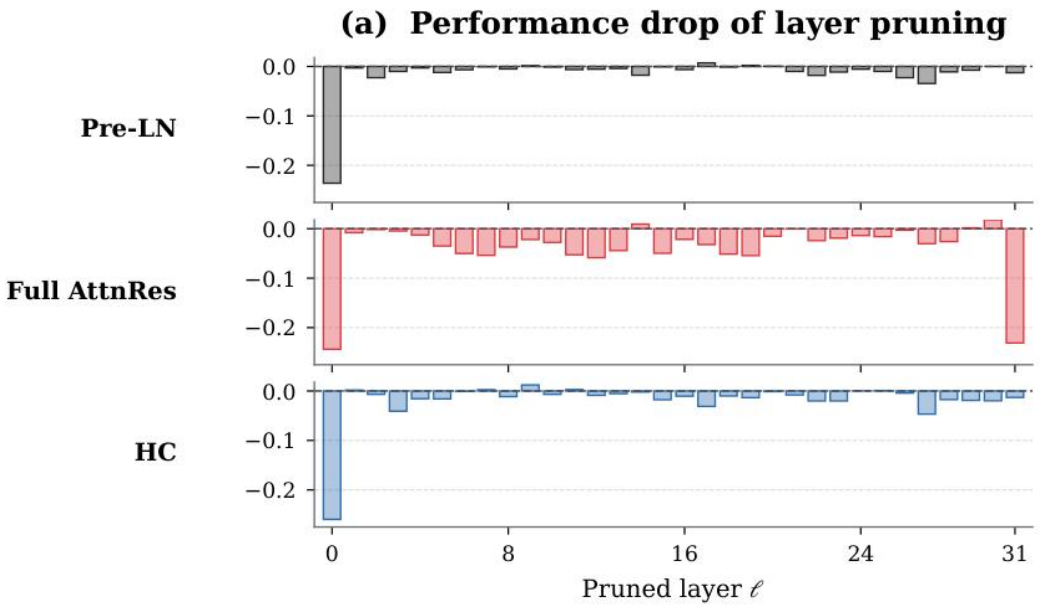
$$P(\ell_1, \ell_2) = \frac{|\mathcal{L}(M) - \mathcal{L}(M_{\text{swap}(\ell_1,\ell_2)})|}{\mathcal{L}(M)}$$

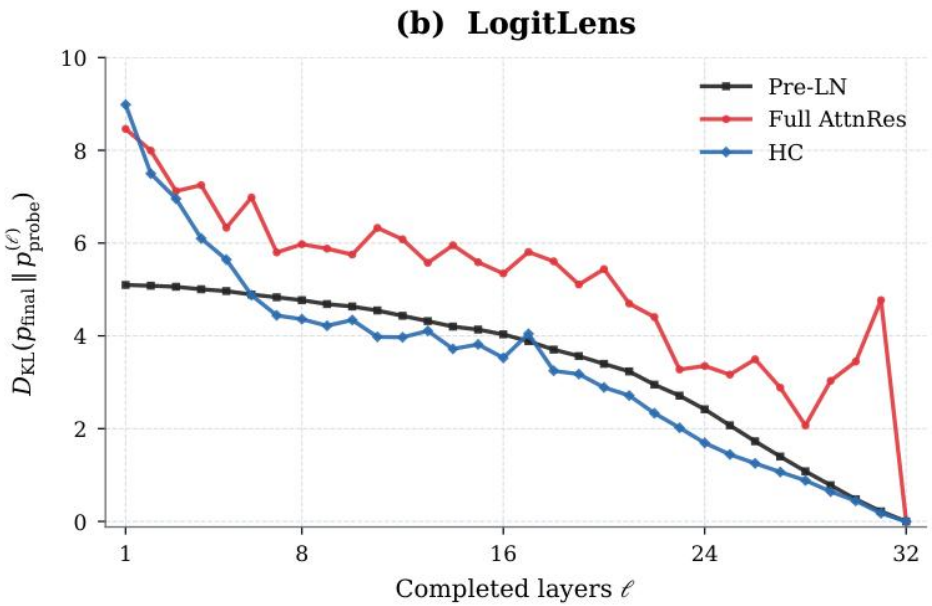


Two routes to effective depth

AttnRes forms a normalized mixture over the embedding and all earlier sublayer outputs, so it **directly retrieves** stored computations.

(m)HC realizes cross-depth access recursively, **propagating and recombining** earlier outputs through interacting residual streams.



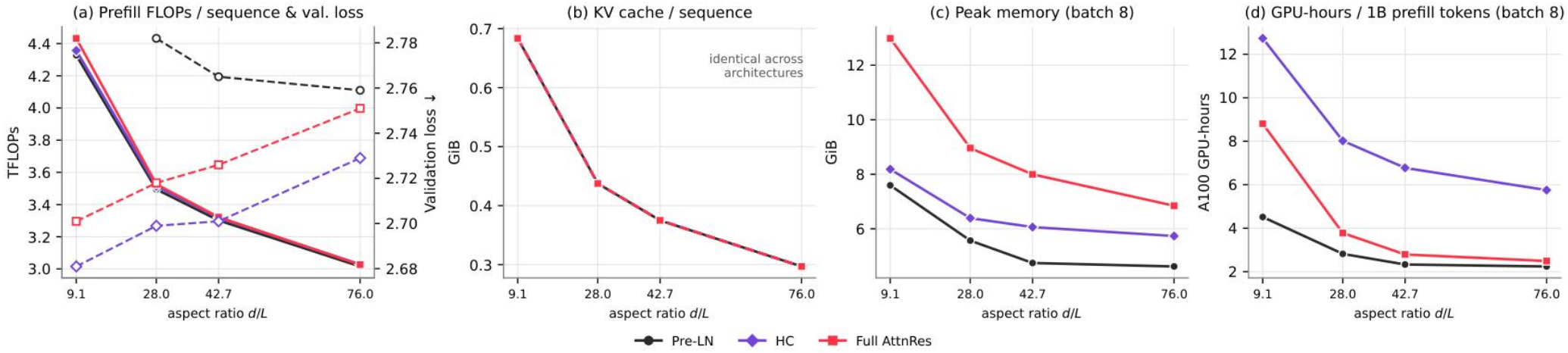


Depth costs systems efficiency

3.0 → 4.4
prefill TFLOPs / sequence, d/L 76 → 9.1

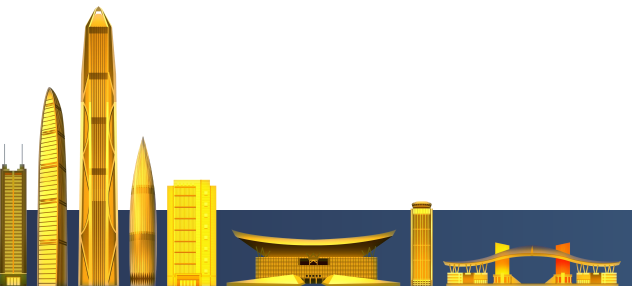
> 2×
KV cache, identical across architectures

Sharp rise
peak memory and GPU-hours at the deepest shapes



Conclusion

- Computational Depth is an underexplored and promising scaling axis — but only with the correct residual connections, and system-level design is badly needed to fulfill it in practice.



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Thanks!

